The Blessing or Curse of Petroleum Exploration in Nigeria –

a critical look at Oil Spillage

GEOG418 Project

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Abstract

Prior to the discovery and subsequent exploration of Crude Oil in Nigeria in 1956, Agriculture was the mainstay of Nigeria's economy through which, cash crops like; palm produce (oil and Kernel), cocoa, rubber, timber and, Groundnut etc. were exported, thus making Nigeria a major exporter in that respect. Indeed, agriculture offered vast opportunities and employed over seventy percent (70%) of the Nigerian labour force. Crude oil took over as the main foreign exchange earner and with the rising prices of the product over the years, one would be forgiven to assume that Nigeria would have blossomed and transformed from an emerging market into a developed nation by now.

This work looks critically at petroleum exploration as it has affected the oil producing states of the Niger Delta (Rivers State). It looks at how the lives of the inhabitants have been affected and if they actually think of the resources from their region as a blessing. The study is important to make government aware of the plight of the people, environmental pollution, diversification of the economy and for this study, I will focus on Rivers State and I would employ both primary and secondary data sources in this study.

Introduction

Nigeria is the largest oil producer in Africa and the 13th in the world. In fact, at some point Nigeria was the 7th largest oil producer with over 2 million barrels produced in a day. As at 1976, Nigeria had already hit the 2million barrel mark and one would have been forgiven to imagine a land of "milk and honey" as they like to be called. Indeed the early years showed promise with huge earnings from the petroleum industry which reached a peak during the first gulf war. Nigeria's oil earnings hit an all-time high. Over the last seventeen years, Nigeria has recorded huge earnings from the petroleum industry. Data compiled from the Central Bank of Nigeria, CBN, showed that Nigeria earned N77.348 trillion from the oil and gas industry from 1999 to 2016. Analysis of the various oil and gas earnings showed that after various deductions, net oil revenue over the same period stood at N41.038 trillion. A breakdown of the earnings showed that from 1999 to 2003, the country's gross earnings stood at N7.329 trillion, while net oil revenue stood at N4.713 trillion; gross oil revenue and net oil revenue from 2004 to 2007 stood at N17.868 trillion and N9.561 trillion respectively.

Ironically however, fast forward to 2017 and billions of dollars later in investments to improve the industry, Nigeria still produces just over 2 million barrels a day and despite the huge resources earned from petroleum, the country has nothing concrete to show. Nigeria is still besieged with inadequate infrastructure, epileptic power situation, low foreign exchange reserves, low savings and an abysmally low standard of living. The country's currency is currently exchanged at about N360 to a dollar. Nigeria's gross oil earnings translates to an average annual earning of N4.55 trillion, while the net oil revenue translates to average annual earning of N2.414 trillion. Nigeria's net oil revenue of N41.038 trillion would have been able to build nine refineries with a refining capacity of 650,000 barrels per day refineries at N4.32 trillion each, about $12 billion. It would have helped increased Nigeria's refining capacity by 6.5 million barrels per day helping the country effectively end fuel importation. (www.vanguardngr.com).

The Study Area

There are nine oil producing states in Nigeria but for this study, I have chosen Rivers State because it has suffered some of the worst cases of neglect and environmental pollution. Home to the garden city – Port Harcourt, Rivers state was one of the most beautiful states in Nigeria. Rivers state is located in the southernmost part of the country and has a population of more than 5 million people, making it the sixthmost populous state according to census data released in 2006. Its capital, Port Harcourt is the largest city, and is economically significant as the centre of Nigeria's oil industry. Please see the study area below.

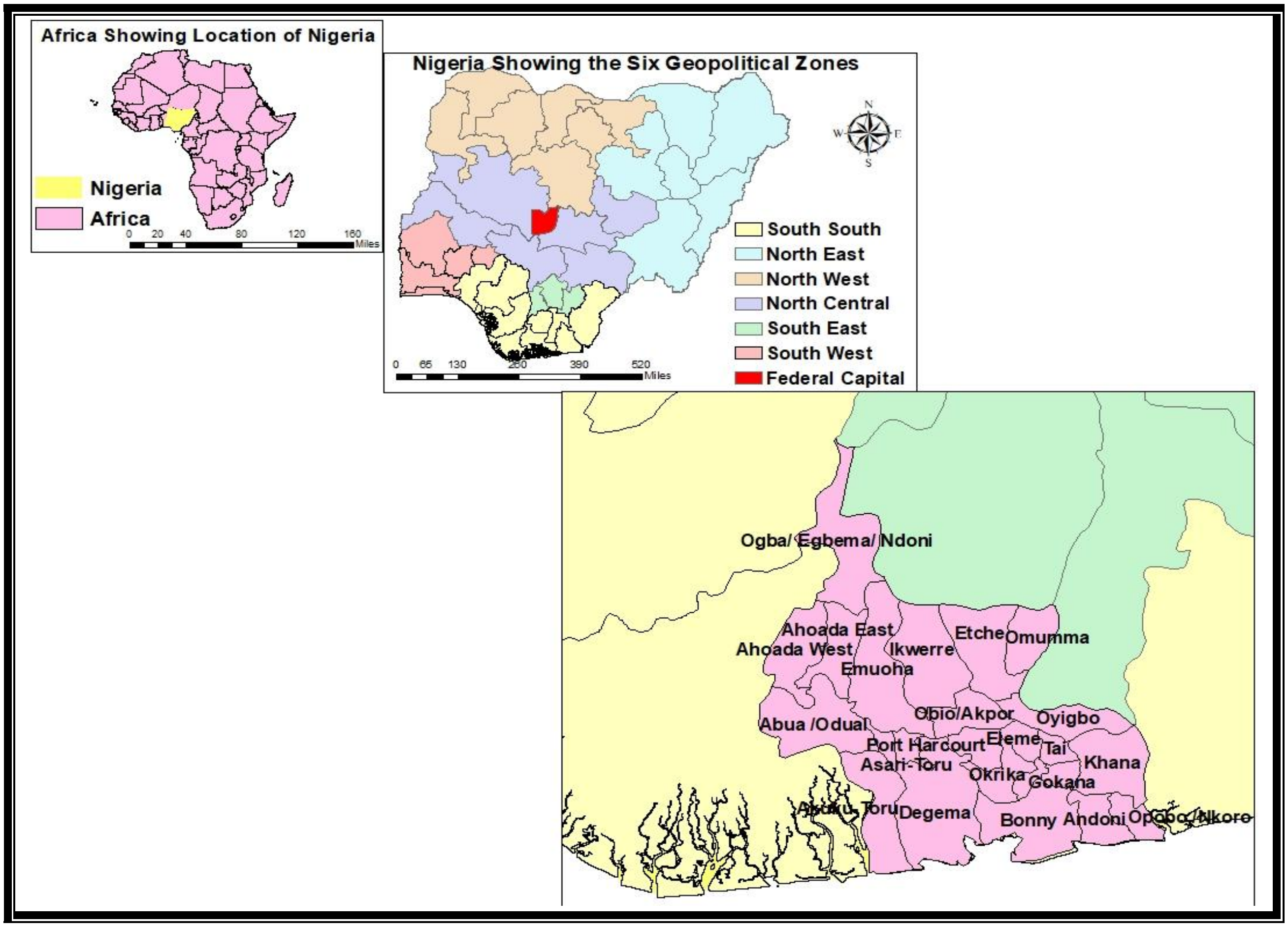

Fig.1 showing the study area: Rivers State.

Research Objectives

The specific objective of this study is to consider the gains of petroleum exploration, the huge foreign exchange earnings, sitting at the table of oil producers and pit these against the standard of living of the people in the oil producing communities, examine the environmental situation and allow the reader draw conclusions as to whether the discovery and subsequent exploration of crude oil has been a blessing to Nigeria or has indeed been a curse.

The broad objectives are;
1.To identify other key dependent variables associated with the topic
2.To measure the effect of each determinant.

The Niger Delta covers 20 000 km2 within wetlands of 70 000km2 formed primarily by sediment deposition making up 7.5% of Nigeria's land mass *(academia.edu).* It is home to 20 million people and over 40 different ethnic groups. Since 1956 when crude oil was discovered in Nigeria, the region has constantly been in the news, both for good and bad reasons. The region became the darling of everyone (home and abroad) because of the vast resources of crude deposits and the environmental issues which eventually came to light as a result of the exploration. The key environmental issues in the Niger Delta of Nigeria relate to its petroleum industry. There are three main issues of pollution as it concerns the petroleum industry namely:

- Oil Spillage which affects both land and water.
- Air pollution from gas flares, chemical emissions and radiation.
- Sound: from the equipment used.

All these problems, the people of the region are exposed to daily and the socio-environmental consequences of oil exploration have been reported internationally (Finer, Jenkins, Pimm, Keane, & Ross, 2008; Ikporukpo, 1983), however, not until about a decade ago did the issue of the environmental pollution come to light. All we heard of was the gains of oil production and the problem was announced because the social unrest problem reared its ugly head. Of these however, oil spillage is the most critical as virtually everyone is affected. The main occupations of people in the Delta is Fishing, Farming and Hunting and most of their land has been taken over either by government because of the presence of oil or by oil spillage which ravages the area making any form of activity impossible.

Figure 1 below shows a typical oil spill site in the delta, oil spills and flows all over the land thereby affecting commercial activities and destroying the organisms in the soil, aquatic fauna and displacing people and animals from their habitats.

Fig. 2: showing a typical oil spill site.

Extent of the problem

Reports on the extent of the oil spills vary. The Department of Petroleum Resources estimated 1.89 million barrels of petroleum were spilled into the Niger Delta between 1976 and 1996 out of a total of 2.4 million barrels spilled in 4,835 incidents. (approximately 220 thousand cubic metres). A UNDP report states that there have been a total of 6,817 oil spills between 1976 and 2001, which account for a loss of three million barrels of oil, of which more than 70% was not recovered. 69% of these spills occurred off-shore, a quarter was in swamps and 6% spilled on land.

The Nigerian National Petroleum Corporation places the quantity of petroleum jettisoned into the environment yearly at 2,300 cubic metres with an average of 300 individual spills annually. However, because this amount does not take into account "minor" spills, the World Bank argues that the true quantity of petroleum spilled into the environment could be as much as ten times the officially claimed amount. The largest individual spills include the blowout of a Texaco offshore station which in 1980 dumped an estimated 400,000 barrels (64,000 m^3) of crude oil into the Gulf of Guinea and Royal Dutch Shell's Forcados Terminal tank failure which produced a spillage estimated at 580,000 barrels (92,000 m^3). In 2010 Baird reported that between 9 million and 13 million barrels have been spilled in the Niger Delta since 1958. One source even calculates that the total amount of petroleum in barrels spilled between 1960 and 1997 is upwards of 100 million barrels (16,000,000 m^3). (nnpcgroup.com).

So in essence, the true extent of the spills is unknown and there is a lot of dancing around the main issues. The oil companies are dragging their feet, government is not enforcing laws, all we see is huge amounts budgeted for clean-up in elaborate ceremonies with nothing being actually done on ground.

Methods & Data Acquisition

I employed primary and secondary methods of data collection for this study using existing research and data, collecting data from direct sources. I used geographical methods, GIS to analyze data collected to form a clearer picture of the geographical locations, the population statistics and the impact of oil exploration which will were clearly represented on the maps and bar charts. Analysis of population density was carried out using secondary data obtained from population data from *citypopulation.de*, I obtained data of oil spills for Shell from the Department of Petroleum Resources. Hotspot analysis was carried out to show the areas with the highest impact of oil spills, effect of oil spills on land use analysis was also carried out in relation to the oil spill sites. Data for land use was obtained from globallandcover.com while data for location of villages was obtained from geofabric.de. For the land use, land cover data I received had more than 20 classes which I reclassified to show just Forests, Agriculture, Water bodies and Grassland.

The data was suitable for this study as they depicted exactly what I wanted to highlight and was compatible with ArcMap able to manipulate them properly.

Data Methodology

I employed ArcMap 10.5.1 to complete the GIS tasks of this study in its entirety. It was a very valuable tool in clipping Rivers State by attribute when I created an inset map to show the study area. I also used this when I created the land use map for Rivers State. I obtained the oil spill data for the months of January for seven years (2011-2017), converted these to a single unit, decimal degrees and brought into ArcMap as X, Y data. It was very interesting to see the spill points on the map. Using the population data gathered, I created a population density thematic map and brought in the oil spill points to show exactly what section of the population was affected by the spills. For more emphasis, I created a density release of the spill points to show what areas were the worst affected and a 3 by 5 by 7 mile buffer around the oil spill sites and made it transparent so it can be viewed in relation to the area. I converted the density map to Raster and used the Hillshade effect and stretch to enhance the effect. I also emphasized on a map the poverty levels of people in Rivers State, showing how the spills affected people according to their ability to meet their basic needs.

I carried out the same process using the land use map after reclassifying it into four (4) units Forest, Agriculture, Water bodies and Grassland, I brought in the oil spills to show how it affect the occupations of the people. I also created a 3 by 5 by 7 mile buffer for emphasis. For my final maps, I carried out a statistical analysis of hotspots in the state dividing the number of villages by the number of oil spills. To successfully achieve this, I also carried out an Inverse Distance Weighting (IDW) of the results.

Results

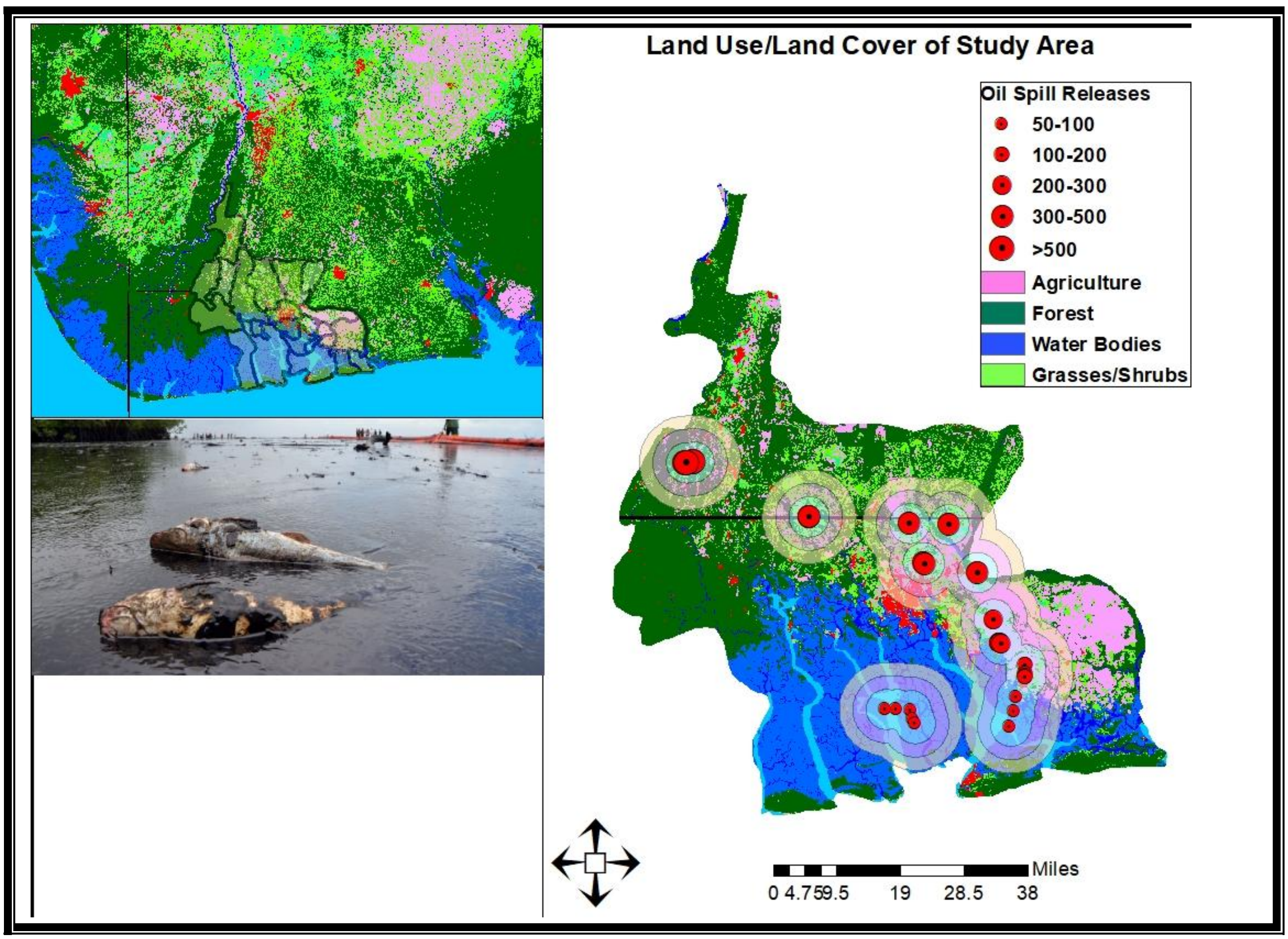

Figure 3**. Showing how and use is affected by spills. Notice the effect over Land and Water.**

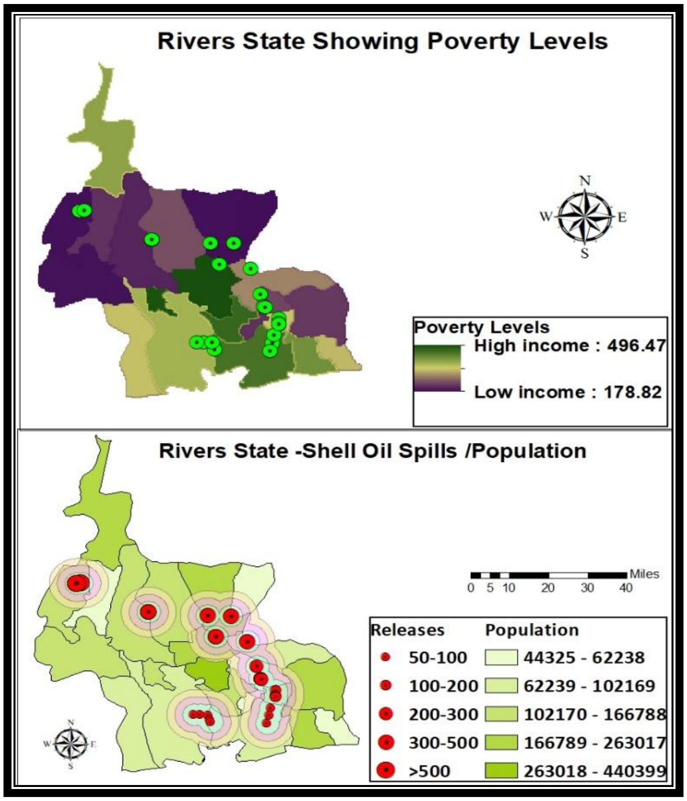

Figure 3 showing the map of poverty levels & Population density in Rivers state against oil spill incidents.

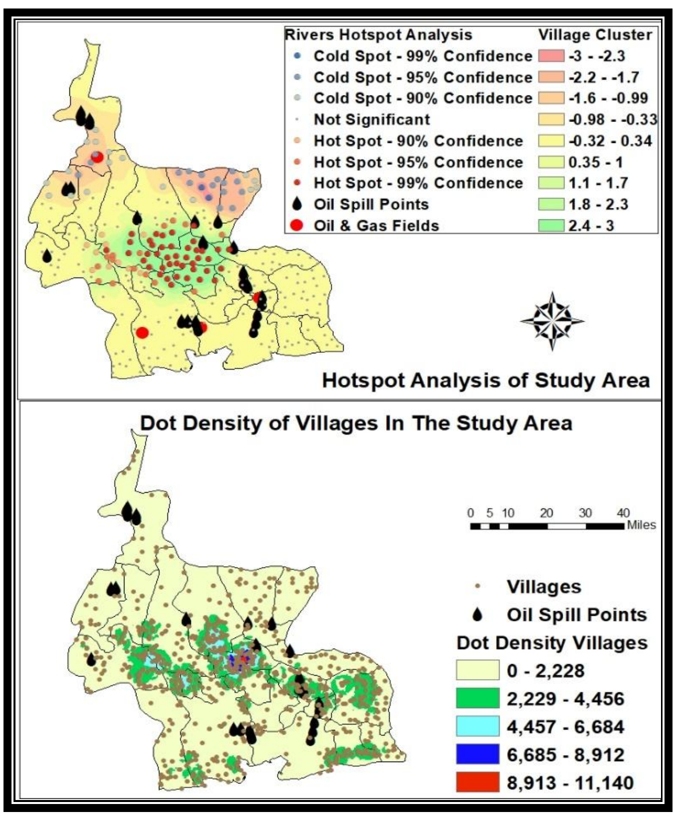

Figure 4 shows the result of the hotspot analysis and the dot density map of villages.

Figures 5, 6 & 7 depicting the state of the Niger Delta today.

S/N	DESCRIPTIONN&	DIMENSIONS	DATE TAKEN
Fig 5	Depicts what used to be a farmland taken over by oil spills. Sights like these are very common in the delta.		March 2007
Fig 6	Violence and kidnappings are now rampant in the delta. Youths have taken up arms against government in a struggle to control "their resources"		January2011
Fig 7	The youths of the delta have resorted to violence and oil "bunkering" meaning they steal the oil to resell at cheaper rates. Since government has turned a blind eye to their plight, they have resolved to take care of themselves and their families by doing this.	750 x 499	December 3, 2009

<u>Discussion</u>

I wanted to display the quantity of oil spills per area but for lack of space. I wanted to compare it against the dot density and the hotspot analysis because the results of those prove the validity of the analyses in the sense that the region with the highest density of population also has a high concentration of villages and is the area with the highest confidence of 95% in the hotspot analysis. It is also interesting to note that the most densely populated areas have high occurrences of oil spills. The results of the land use analysis shows just how devastating oil spills have been on the Niger Delta. Every facet of life is affected, land and aquatic habitats are destroyed. The figure above shows just a little bit of the effect of oil spills on the area.

Putting the figures together, a total area of 655 225 m^2 was affected just from one company and for just a particular month over a period of time. Knowledge of this helps is forming a picture of the effect. I also wanted to look at the effect on Social activities. The delta has become a war zone with kidnappings, violent clashes between youths and the military and armed robbery. As a result of poverty, most of the youths have resorted to violence and robbery to make ends meet.

The Niger Delta is ravaged by oil spill in the following ways:

E**conomically**- Corruption and huge losses due to reduced production. In almost thirty (30) years and billions of dollars spent, Nigeria has remained on the 2 million barrel per day mark. There is very little to show for the years of sale of crude oil but poverty, violence and displacement.

Environmentally- the once fertile areas have been turned into barren land, water pollution and soil degradation means aquatic fauna and hunting which provide a source of livelihood for the inhabitants are destroyed. People are driven from their homes with little or no compensation.

Socially - Social unrest due to competition for resources by government and militants. From an area of splendor, the delta has become synonymous with violence. There are now fixed military bases and it is not uncommon to experience clashes between government forces and the youths who have been armed.

Conclusion

How long will this go on?

Only in 2008 when more than 600 000 barrels was spilled by shell did the cry of the people and humanitarian agencies reach the ears of government and since then, it’s been promises of clean up with nothing concrete taking place. Saudi Arabia, United States of America, even Angola right there in Africa are oil producing countries that have turned crude oil as a resource into a blessing, improving the lives of their people. Why can’t the same happen in Nigeria?

On the one hand there is the commodity that brings immense resources to Nigeria but on the other, untold suffering and neglect to the people for over 50years. So those living in the Niger Delta can call it a “curse” but for the rest of the country just reaping the benefits, it might be called a blessing. It would be fool hardy to suggest that government cease all exploration to focus on the clean-up, especially as Nigeria runs a mono-economy that is totally dependent on crude for foreign exchange earning therefore Government must find a way to strike a balance, to ensure the continued existence of host communities, ensuring strict cleanup policies with strident penalties to defaulters. This would go a long way in ameliorating the conditions of the people of the region and bring them out of poverty.

I was able to highlight a problem faced by people in the Niger Delta by asking a geographical question and using geographical methods to answer them or at least further highlight the problems and seek solutions to them. These are the type of questions I hope to ask and seek answers to as I continue to go further in the discipline, using it as a tool to ask questions about the things that affect people in society and use it to also seek answers to those problems. The importance of geography in modern life cannot be over emphasized as

References here

http://www.shell.com.ng/sustainability/environment/oil-spills/march-2016.html
https://www.vanguardngr.com/2017/09/nigeria-earned-squandered-n77trn-oil-revenue-17-years/
https://www.citypopulation.de/php/nigeria-admin.php?adm1id=NGA033
http://globallandcover.com/document/globemapsheet.zip
http://download.geofabrik.de/africa/nigeria-latest-free.shp.zip